

A sampling distribution

A sampling distribution describes a statistic across repeated samples of the same size.

What would one point represent in a distribution of sample means?

Data or sample means?

Imagine 1,000 independent random samples of 25 package masses, with one mean calculated per sample.

Would a histogram of those means display 25,000 masses or 1,000 means?

An observation and an average

X represents one observation; $\bar{X}$ represents the mean of n observations.

Which variable describes the average mass of 25 packages?

Center of the sample means

For independent observations from one population, the mean of the sample mean equals the population mean.

$$E(\bar{X}) = \mu$$

If the population mean is 500 g, where is the sampling distribution centered?

Standard error

The standard error of a statistic is the standard deviation of its sampling distribution.

Does standard error describe variability among individual masses or among sample means?

Standard error of the mean

For n independent observations with common population SD σ , the standard error is σ divided by the square root of n .

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

For $\sigma=20$ g and $n=25$, what is the standard error?

SD versus standard error

Model: package masses have SD 20 g.
Independent samples contain 25 packages.

Which spread belongs to one mass, and which belongs to the sample mean?

Changing sample size

Model: population SD is 20 g. Compare independent samples of $n=25$ and $n=100$.

How does the standard error change?

Reducing standard error

Model: independent sampling gives standard error 4 g at $n=25$.

What sample size would reduce standard error to 2 g?

The central limit theorem

For independent, identically distributed observations with finite variance, sample means become approximately normal as sample size grows.

Does this theorem say that the individual observations become normal?

An exactly normal population

The mean of independent normal observations is normal for every positive sample size.

If masses are normal, do we need $n \geq 30$ to give their sample mean a normal distribution?

A skewed population

Model: individual download durations are strongly right skewed, with finite mean and variance.

Would a sample mean from $n=8$ automatically have an approximately normal distribution?

No universal cutoff

The sample size needed for a useful normal approximation depends on the population shape and the desired accuracy.

Why might $n=30$ be insufficient with strong skewness or extreme outliers?

Independence

The formula $\sigma/\sqrt{n}$ assumes independent observations.

Could 50 measurements taken moments apart from one unstable machine behave like 50 independent observations?

Sampling without replacement

When sampling without replacement, a sample no larger than 10% of the population is a common independence approximation.

Would $n=200$ from a population of 500 comfortably meet that guideline?

A sampling model

Model: independent masses follow $N(500, 20)$, measured in grams. Samples contain 25 packages.

What normal model describes the sample mean?

Standardizing an average

A sample mean is standardized using its standard error.

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

For $\mu=500$, $\sigma=20$, and $n=25$, what z-score belongs to $\bar{X}=508$?

An upper-tail average

Model: independent masses follow $N(500, 20)$.
Samples contain 25 packages.

What is $P(\bar{X} > 508)$?

A lower-tail average

Model: independent masses follow $N(500, 20)$.
Samples contain 25 packages.

What is $P(\bar{X} < 496)$?

An average within a range

Model: independent masses follow $N(500, 20)$.
Samples contain 25 packages.

What is $P(496 < \bar{X} < 504)$?

A percentile of sample means

Model: independent masses follow $N(500, 20)$.
Samples contain 25 packages.

What is the 95th percentile of the sample-mean distribution?

One value versus a mean

Model: independent masses follow $N(500, 20)$.

Which is more likely: one mass above 508 g or a mean of 25 masses above 508 g?

The population remains variable

Model: population SD is 20 g. Increase the sample size from 25 to 100.

Does the spread of individual masses shrink?

A surprising sample mean

Model: independent masses follow $N(500, 20)$.
A synthetic sample of 100 has mean 507 g.

How unusual is that mean under the model?

From probability to inference

A population mean is unknown. A random sample mean is observed.

How could the sampling variability of means help estimate the unknown population mean?